Final Team Report

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EK 210 Section A4, Professor Kia, Fall 2025
Group 2: Deskinator

Executive Summary

This project involved designing, building, and testing the Deskinator, an autonomous robot created to clean debris from a 4’ × 4’ desk surface while also meeting requirements on size, weight, and runtime. The robot is to complete cleaning in under two minutes, operate unsupported for at least five minutes, remain under 12” × 12”, and weigh less than 1500 g.

The final product uses DC gearbox motors for movement and a custom designed vacuum system to collect debris such as rice grains and paper scraps. Infrared sensors monitor the edges of the desk to prevent the robot from falling, and a touchless wave-to-start mechanism activates the robot without requiring any physical contact. An Arduino Uno controls all of the device’s functions, including navigation, sensing, and vacuuming.

Testing showed that the Deskinator successfully collected debris, navigated the desk safely without falling off the edge, and covered a majority of the surface within the time constraint. The robot met the size and runtime requirements and performed well over multiple runs, demonstrating its reliability. Overall, this project shows that a compact and autonomous desk-cleaning robot can be created with inclusivity of all users in mind.

Introduction & Problem Statement

Maintaining clean workspaces in classrooms, laboratories, and study environments is essential for hygiene and productivity. However, manual cleaning is inefficient, inconsistent, and time-consuming. Small debris, such as dust, paper shreds, and food crumbs, can accumulate quickly and be tedious to clean up. Students, professors, and engineers lead busy lives and need time to attend to tasks other than cleaning. This project addresses the need for a compact, automated cleaning solution that operates independently, requiring very little user input, while also cleaning efficiently. The Deskinator device was developed to provide an accessible and reliable approach to autonomous desk cleaning, enabling users to focus on more crucial tasks without worrying about maintenance.

The goal is to design and build an autonomous robot that cleans and contains debris, so users like students, engineers, and Professor Kia can attend to other tasks. To do so, the product must achieve multiple objectives that have guided the design process (Table 1). With a budget of $400, the device must be small and compact, spanning a maximum of 12” × 12” and weighing less than 1500 g or about 3.3 lb. The robot will operate under activation through a touchless wave-to-start system, and then drive safely on its own, detecting edges so as not to fall off the desk surface. This hands-free system makes the Deskinator accessible for users with limited mobility. As the device traverses at least 90% of a 4’ × 4’ desk surface, it collects at least 95% of the debris using a custom vacuum system that also stores the collected debris, all under 2 minutes. Once the cleaning job is completed, the robot signals completion visually through an LED light and audibly through a buzzer, ensuring inclusivity for users who are both hearing and visually impaired. The Deskinator is able to operate for 5 minutes without requiring any external support. These objectives guide the development of the project and ensure that the final system balances cleaning performance, mobility, power efficiency, safety, and accessibility.

Design Elements

The Deskinator's design was motivated by the need to maximize cleaning efficiency and reliability, while still falling within the constraints of size, weight, and power. The Deskinator consists of two main subsystems: a vacuum system that suctions and stores debris, and a motion system that moves the robot safely across the desk surface, detecting edges to prevent it from falling off.

To maximize debris collection, a brushless DC centrifugal fan was incorporated into a custom vacuum design. Testing of nozzle sizes determined this configuration to be the most effective, as a nozzle of about 11.5 mm diameter was capable of suctioning the most rice grains (Figure 1). Right behind the nozzle is a line of brush bristles to sweep the rice and debris forward into the nozzle to be suctioned. At the front of the device are sponges that act as a barrier to sweep the debris toward the center, where the vacuum nozzle is.

The motion system includes 3 wheels. DC gearbox motors were chosen to drive the robot's two front wheels due to their simplicity and sufficient torque. These motors provide consistent forward motion with the ability to turn. The third back wheel is an omnidirectional caster wheel that provides steering and stability. An IR distance sensor was used for navigation and edge detection (Figure 2). The sensor was attached to an arm that pivots using an S51 Micro Servo Motor. This sensing arm rotates 180° to detect where there is an edge. If an edge is detected, the robot will turn right and continue driving forward until it approaches another edge. This sequence continues until the robot traverses the entire desk surface.

The entire robot starts with a PIR motion sensor on its top face (Table 2). When the user waves their hand over the robot, the PIR sensor will detect this motion as a start signal, and a green LED turns on to indicate that the device has started working. When cleaning has been completed, a red LED turns on to indicate that the device has stopped. The entire robot is powered through a 12 V battery (Figure 3).

Evaluation of Results

The performance of the Deskinator was evaluated against the project objectives previously defined, with an emphasis on effective cleaning, reliability, and effective navigation (Table 3). Testing demonstrated that the final product met most of the important functional requirements while also recognizing areas for future improvement.

The Deskinator successfully collects lightweight debris, such as rice and paper shreds, showing that the vacuum suctioning system is effective. Visual inspection of the debris container showed that the vacuum collects a consistent amount of debris across all testing trials, adding an aspect of reliability to the vacuum system. During testing, the robot consistently traversed a majority of the 4' × 4' desk surface. The sensing and navigation system also successfully detected the edges of the desk, preventing the robot from falling off and only driving within the bounds of the desk surface. The robot completed multiple trials without falling off the table, ensuring reliability.

The Deskinator consistently completed full cleaning cycles within the provided two-minute time limit. The battery performance exceeded the minimum five-minute runtime requirement. Physical constraints were also met, as the robot body ended up being 9.25” × 9.5”, falling within the maximum size of 12” × 12” (Figure 4). The final product also stayed under the $400 budget for the project (Table 4).

However, despite meeting most objectives, some requirements were unable to be achieved. Because of all of the components that had to go into the robot, the robot was unable to remain under the weight limit of 1500 g. It ended up being 1638 g, slightly above the weight limit. Still, it was determined during the Problem Definition phase that the device being lightweight was not a primary objective, and that other objectives, such as effective cleaning, reliability, and effective navigation, were more important and needed more focus. The device falling short in this aspect and not meeting all objectives shows that design trade-offs are sometimes necessary to prioritize the most crucial functions. It also shows that there is always room for improvement and that all aspects of the design should be considered for the future.

Alternative Design Choices & Lessons Learned

There were many various design options considered during the Conceptual Design phase (Table 5). Due to a constrained time period, certain components in the initial design were not incorporated into the final design. For instance, the front roller was not incorporated due to multiple motor failures. In order to save time to work on critical components, the roller was scraped. The housing of the robot could be adjusted to allow easier accessibility to the electronic components. Regarding the power source, the battery could be lighter to meet the overall constraints of this design project. Instead of a singular sensor on an arm attached to the servo motor, multiple distance sensors could be attached to the housing to detect a table top edge. The current touchless start of the robot includes a PIR motion sensor. For future iterations of this project, the electronic device could be switched to a sound sensor. The user could create any loud sound to activate the device.

References

Sharp Corporation, “GP2Y0A21YK0F Distance Measuring Sensor Datasheet,” Sharp Electronics, n.d. https://global.sharp/products/device/lineup/data/pdf/datasheet/gp2y0a21yk_e.pdf

“L298n Motor driver Arduino | Motors | Motor driver L298n,” Arduino Project Hub, Jun. 12, 2021. https://projecthub.arduino.cc/lakshyajhalani56/l298n-motor-driver-arduino-motors-motor-driver-l298n-7e1b3b

OpenAI, “ChatGPT Used for drafting assistance,” 2025.

Appendix

Table 1: Objectives and their corresponding metrics ranked from most important to least important

Rank	Objective	Metric/Specification
1	Clean Effectively	Picks up >95% of debris that passes by the cleaning mechanism
2	Be Reliable	Runs multiple (>5) times without falling off the table or failing to clean the surface
3	Navigate Effectively	Covers >90% of the surface
4	Fast Cleaning	Cleans surface in under 2 minutes
5	Have an Effective Runtime	Operates for 5 minutes without external support
6	Lightweight	≤ 750g (1.65lbs)
7	Compact Footprint	≤ 12” x 12”

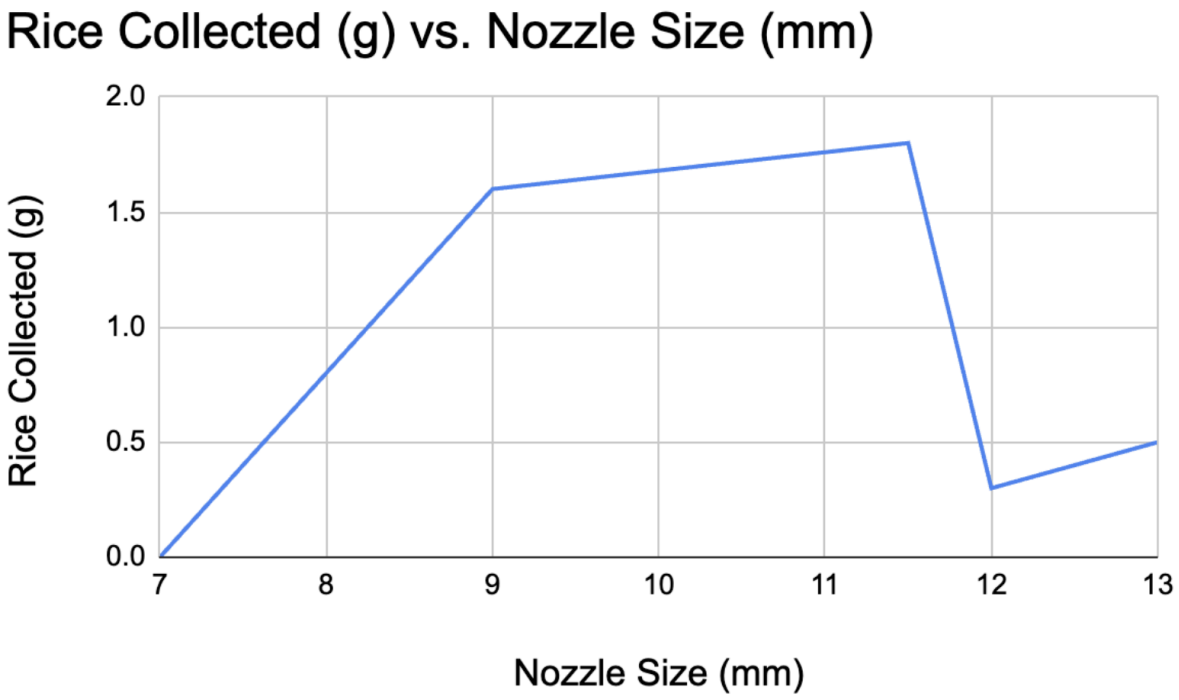


Figure 1: Graph of mass of rice collected for various vacuum nozzle sizes

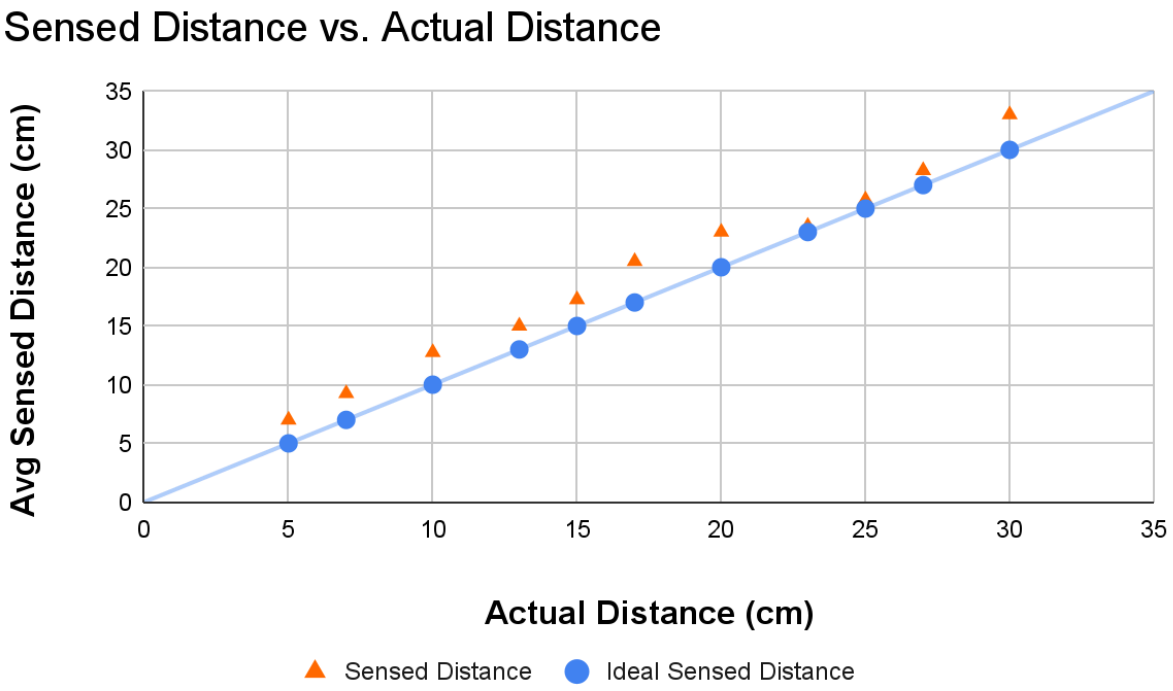


Figure 2: Graph of IR Distance Sensor accuracy

Table 2: PIR Motion Sensor Response Time

Trial #	Detection latency (seconds)
Trial 1	1.2
Trial 2	1.5
Trial 3	1.3
Trial 4	1.4
Trial 5	1.3

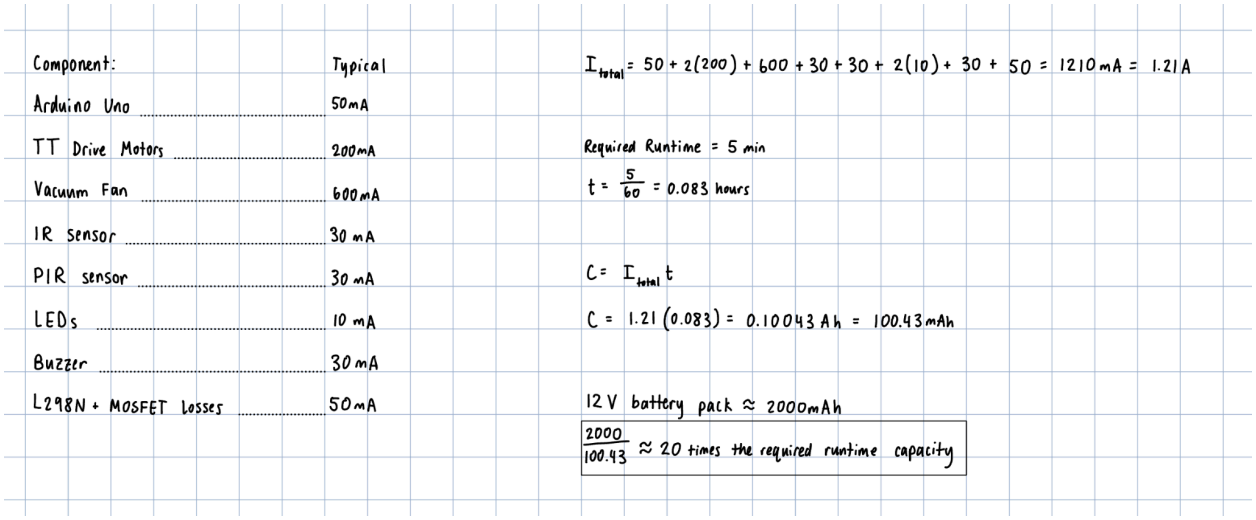


Figure 3: Power Budget Calculations

Table 3: PCC ranking of the importance of objectives

If a row is more important than a column put a 1, if a column is more important than a row put a 0									
	Goals	Size	Runtime	Lightweight	Fast Cleaning	Reliability (falling)	Effective Cleaning	Effective Navigation	Score
	Size	-	0	0	0	0	0	0	0
	Runtime	1	-	1	0	0	0	0	2
	Lightweight	1	0	-	0	0	0	0	1
	Fast Cleaning	1	1	1	-	0	0	0	3
	Reliability (falling)	1	1	1	1	-	0	1	5
	Effective Cleaning	1	1	1	1	1	-	1	6
	Effective Navigation	1	1	1	1	0	0	-	4

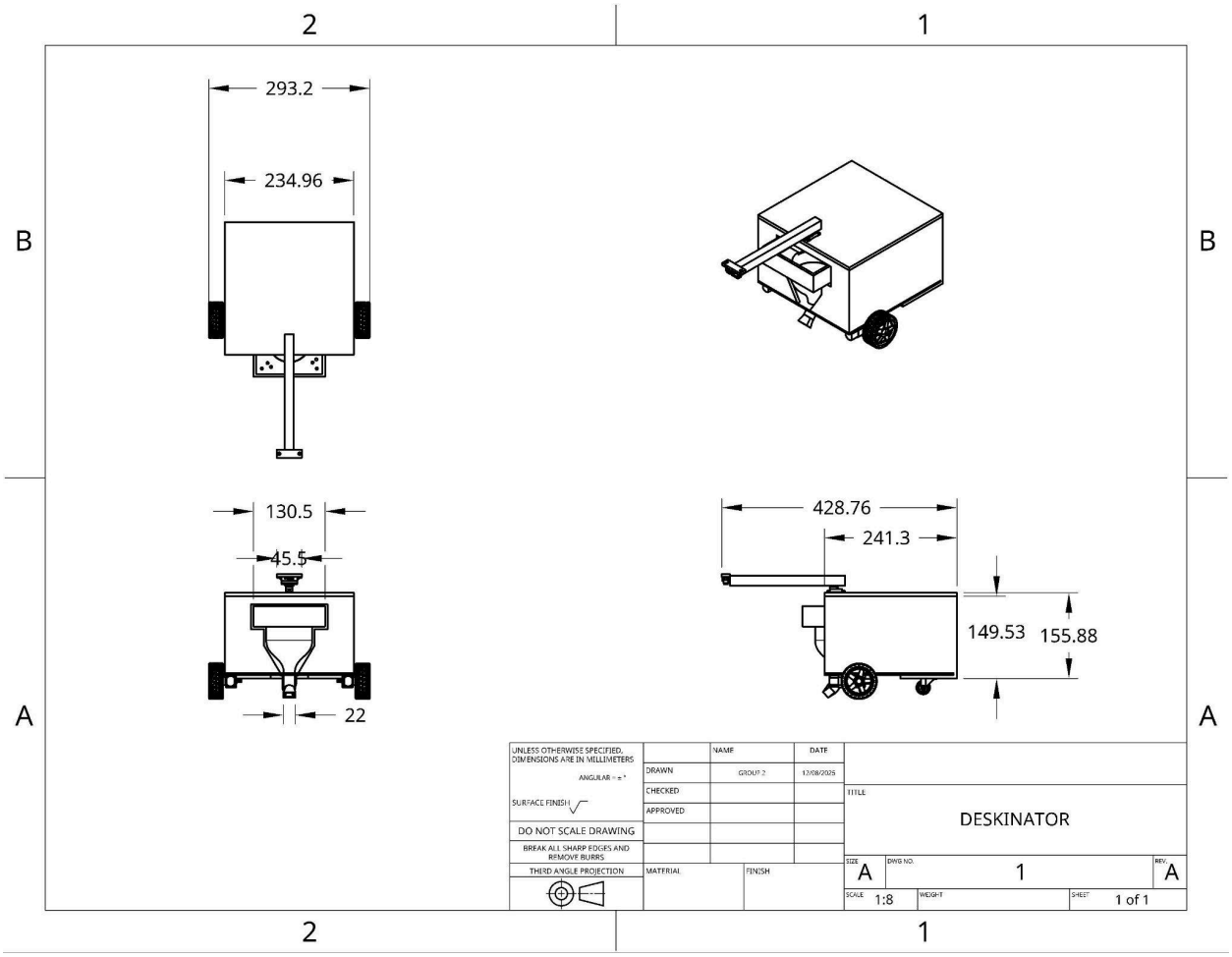


Figure 4: Final Assembly CAD Drawing

Table 4: Bill of Materials with costs of each part

Part name	Part number	Description	Qty	Cost / Unit	Total Cost
DC Motor 3V - 12V	B0BR7S1DQZ	TT Motor 200RPM	1	7.99	7.99
Wheels	B0BR7S1DQZ	Dia: 50mm	0	0.00	0.00
Motor Driver	L298N	Dual H Bridge Motor Drive Module	0	0.00	0.00
IR Distance Sensor	SHARP GP2Y0A21YK0F	range 4cm to 30cm	1	6.30	6.30
Micro Servo Motor	X001YZNRXN	S51 Micro Servo Motor	1	2.00	2.00
Buck Converter	B07XXWQ49N	DC Buck Converter Module	1	7.50	7.50
PIR Sensor	HC-SR501	Range 5M	1	1.99	1.99
DC Motor 12V	RM-ESMO-05E	136RPM Worm N20 High Torque	1	9.30	9.30
Caster wheel	B07GLPJJ7D	Dia: 25mm	1	1.12	1.12
DC Centrifugal Fan 12V	B07RP14KML	75mm x 30mm	1	8.99	8.99

Table 5: Morph Chart of various means for each function

Function	Option 1	Option 2	Option 3
Cleaning Mechanism	Rotating brush	Suction (mini vacuum)	Sticky roller
Rice Storage	Clear pod case top	Clear pod case bottom	Clear pod case side
Movement	Wheels	Omniwheels	Treads
Edge Detection	Ultrasonic Distance Detector	Spring-powered, sticks out in front,	Wheel on the side
Signals User	Blinking LED	Buzzer	LED + Buzzer
Form	Square	Circle	Rectangle

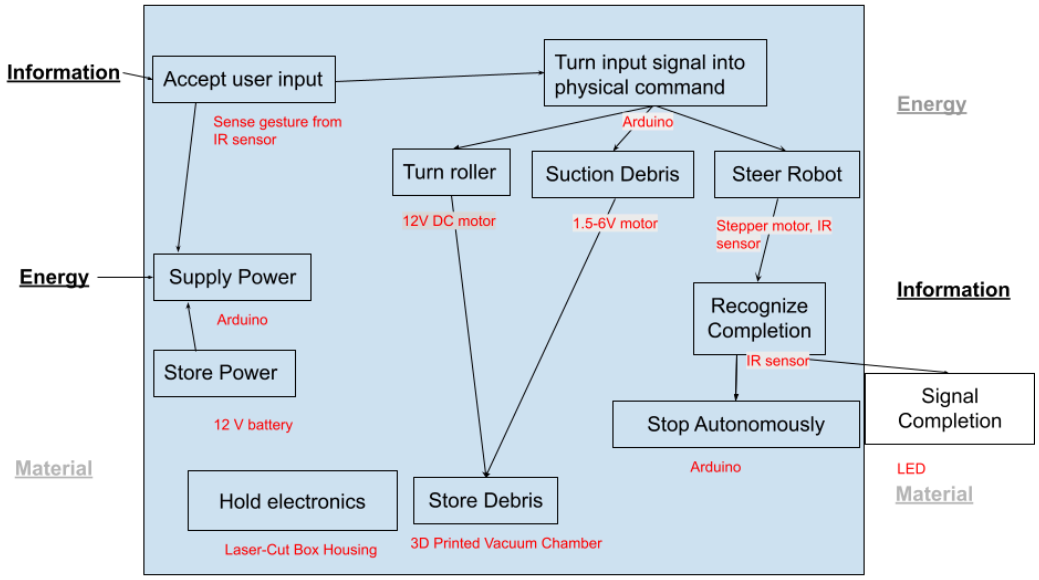


Figure 5: Glass box diagram updated for detailed design

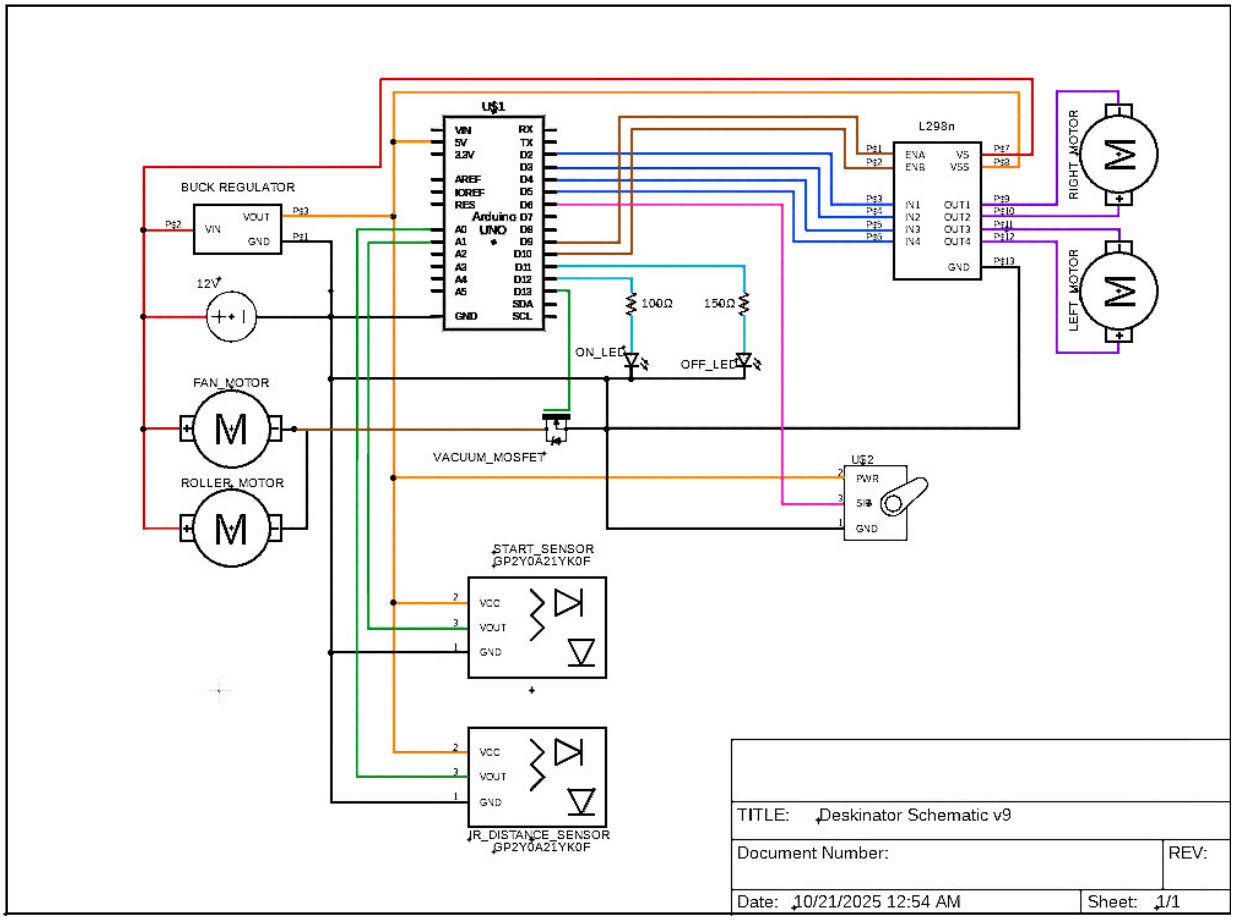


Figure 6: Wiring Diagram

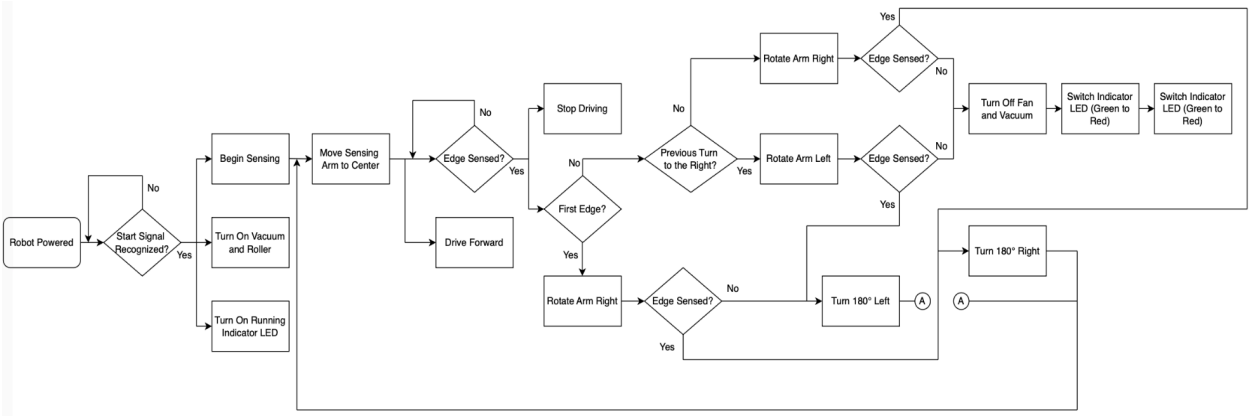


Figure 7: Code Flow Chart